Dr. Elena Rostova

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Academic Summary

Postdoctoral Fellow in Computer Science specializing in Distributed Consensus and High-Throughput Replication.

Education

Ph.D. in Computer Science | Stanford University | 2019 - 2024

Dissertation: Byzantine Fault Tolerant Consensus Protocols for High-Latency Wide-Area Networks

Master of Science in Applied Mathematics | MIT | 2017 - 2019

Bachelor of Science in Computer Science | ETH Zurich | 2013 - 2017

Research Experience

Postdoctoral Research Fellow | Stanford Distributed Systems Lab | Sep 2024 - Present

• Investigated asynchronous state machine replication protocols scaling to 5,000 distributed nodes.

• Formally verified consensus invariants using TLA+ and discovered 3 edge cases in prior literature.

Skills & Technical Competencies

Skills: Distributed Systems, Formal Verification, Rust, C++, Python, Linux, TLA+, Paxos, Raft, Docker, PostgreSQL

Selected Publications & Projects

• E. Rostova et al., "Adaptive Quorum Protocols for Global Consensus", ACM SOSP, 2023.

• E. Rostova et al., "Formally Verified Asynchronous Replication in Rust", IEEE S&P, 2022.